All (Mayoral) Politics Is Local?

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One of the defining characteristics of modern politics in the United States is the increasing nationalization of elite- and voter-level behavior. Relying on measures of electoral vote shares, previous research has found evidence indicating a significant amount of state-level nationalization. Using an alternative source of data—the political rhetoric used by mayors, state governors, and members of Congress on Twitter—we examine and compare the amount of between-office nationalization throughout the federal system. We find that gubernatorial rhetoric closely matches that of members of Congress, but that there are substantial differences in the topics and content of mayoral speech. These results suggest that, on average, American mayors have largely remained focused on their local mandate. More broadly, our findings suggest a limit to which American politics has become nationalized—in some cases, all politics remains local.

At the heart of the republican ideal in the United States is the notion that elected officials properly represent those to whom they are electorally accountable. That is, elected officials in a well-functioning democratic setting are expected to articulate the preferences and desires of the governed. Numerous studies of democratic representation compare some measure of aggregate constituency preferences with some measure of elite behavior or outcomes (see, e.g., Esaiasson and Wlezien 2016). Given the nature of American federalism, this implies that local officials should be concerned with representing and articulating the preferences of citizens at the local level, while federal officials should be comparatively more concerned with national-level policies.

However, recent scholarship suggests that all levels of American politics have become increasingly nationalized. Evidence for such a nationalization effect has been found in US House, Senate, and gubernatorial elections (Aleman and Kellam 2008; Carson and Sievert 2018; Sievert and McKee 2019). Thus, rather than being contested over local issues that are of importance to each respective constituency, political campaigns are largely focused on national issues (Hopkins 2018). While it is true that local constituencies care about national issues and political competition, the stakes of nationalized local politics are high. Indeed, a nationalized style of local politics often comes at the expense of a focus on the local issues that are typically of greater importance to citizens. With the majority of Americans engaging most frequently with the goods and services provided by local governments (e.g., parks, roads, sanitation service), the nationalization of American elections has the potential to weaken both governmental representation and provision at the most local level.[1]

In this study, we examine the extent to which the nationalization of American politics has affected mayoral and gubernatorial representation. Presiding over the most local forms of

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Replication files are available in the *JOP* Dataverse (https://dataverse.harvard.edu/dataverse/jop). The empirical analysis has been successfully replicated by the *JOP* replication analyst. An online appendix with supplementary material is available at https://doi.org/10.1086/716945.

1. A recent report by the Brookings Institution found that, as of 2017, state and local governments "collected and spent amounts equal to 13.1% and 14.7% of GDP [gross domestic product], respectively." Moreover, the increased transfer of federal funds to state and local governments has given these subnational governments a large role in overseeing the "efficient and effective use of public funds" (Nunn, Parsons, and Shambaugh 2019).

Published online March 18, 2022.

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government within the federal system and managing the government services with which citizens interact most frequently, both mayors and governors have substantial influence over the daily lives of their citizens. Accordingly, understanding whether these elected officials have shifted their focus toward national considerations and away from the needs of their local constituencies is of tremendous importance. To address this question, we depart from typical studies of nationalization by analyzing patterns of elite speech with respect to topic similarity via the Twitter social media platform, rather than election results. Adopting this measurement strategy allows us to both validate existing findings from the literature on the nationalization of American politics and to introduce new metrics that capture differences in the behavior of political elites themselves.

We collect a unique data set of 404,049 tweets from US mayors, 102,670 tweets from governors, and nearly 965,000 tweets from members of Congress. Constructing a novel measure of nationalization, our results show that mayors talk about qualitatively different subjects than members of Congress. Governors, by contrast, are more rhetorically in line with members of Congress. Mayoral offices focus predominantly on local topics. Such a finding complements work suggesting that the polarization and partisan antipathy that is pervasive within the context of national politics is largely absent at the most local levels of government (Jensen et al. 2021). We also present results that indicate that the degree to which mayors' rhetoric resembles that of members of Congress is dependent upon the size of the city over which they govern (consistent with their exposure to national trends) and the number of years they have been in office (consistent with their ability to ignore national pressures). As the population size of a city increases, mayors are more likely to engage in nationalized political rhetoric. More time spent in office, on the other hand, is associated with lower amounts of nationalized rhetoric. These results are robust to the inclusion of individual mayoral traits and political preferences, as well as a series of city-level covariates.

This article proceeds as follows. First, we characterize recent work on the nationalization of American elections and develop a theory linking governors and mayors to nationalized behavior manifested through elite speech. Next, we present a series of results based on tweet analyses suggesting that while gubernatorial rhetoric resembles the type of speech typically found among national politicians, American mayors use rhetoric that is distinct from members of Congress. We proceed to show how mayors' comparatively lower levels of nationalized rhetoric vary as a function of individual mayor- and city-level covariates. Finally, we conclude with a discussion about the implications of our results for democratic governance.

ELECTIONS, ACCOUNTABILITY, AND NATIONALIZATION

While canonical models suggest that voters make rational decisions about candidates within each electoral context (see, e.g., Downs 1957), recent evidence suggests that American political behavior has become increasingly nationalized (Abramowitz and Webster 2016; Hopkins 2018). Nationalization, as typically conceptualized by the literature, refers to one of two things. In one case, nationalization refers to the process whereby voters judge politicians—regardless of the electoral level—by their evaluations of the national parties. In the second case, nationalization occurs when state and local elections are largely fought over national (and usually symbolic) issues. Among other causes, these two different forms of nationalization are thought to occur when parties offer similar candidates across electoral levels or the media market changes in a way that prioritizes national over local news (Hopkins 2018; Martin and McCrain 2019). To the extent that this nationalization phenomenon causes state and local officials to prioritize national interests and the concerns of ideologically motivated donor bases to the exclusion of the local citizenry's needs, as Hopkins (2018) suggests, the nationalization of American politics has the potential to drastically alter the relationship between citizens and their elected officials at the local level.

The existing literature on the growing nationalization of American politics has largely focused on US House and Senate elections (see, e.g., Aleman and Kellam 2008; Carson and Sievert 2018). Jacobson (2015), for instance, notes that the incumbency advantage in American politics, long seen as the source of high reelection rates across the country, has been declining over time, and the explanatory power of partisanship in predicting election outcomes for House elections has increased tremendously. Such a shift indicates that Americans today care less about the specific person who represents them and more about the partisan balance of power in Congress. This implies that, unlike in earlier eras, it is increasingly difficult for politicians to court the "personal vote" in their districts (Fenno 1978; Mayhew 1974).

Outside of federal elections, scholars have focused almost entirely on how the nationalization of American politics has affected gubernatorial elections. Hopkins (2018) shows that the state-level correlation between voting for the Democratic presidential candidate and the Democratic gubernatorial candidate has increased considerably over time. Nearly 30 years ago, the correlation between voting patterns at these two electoral levels was a moderately strong .61. By 2010, the correlation had strengthened to just under .9. Thus, there is an increasingly strong connection between the percentage of the two-party vote accruing to the Democratic presidential candidate and the percentage of the vote received by the Democratic

gubernatorial candidate (see also Sievert and McKee 2019). These relationships are also found in state legislative elections, albeit inconsistently. For instance, Rogers (2016, 220) finds that presidential approval shapes how voters view their state legislators: voters will cast their ballots "for the state legislative challenger instead of the member of the president's party" when they are displeased with the president's job performance. Related work finds that this dynamic of nationalized candidate evaluations is most pronounced when national polarization is high (Zingher and Richman 2018). However, contrary to the phenomenon of the nationalization of American elections, others have found that state legislators are largely responsive to their constituencies (Fouirnaies and Hall 2021) and that one-third of counties in the United States tend to vote for one party for president and another for state legislature (Trounstine 2017).

Mayors may be somewhat immune from these national trends: voters cast ballots in mayoral elections based upon retrospective local economic conditions, such as local unemployment, and this effect typically dwarfs that of national economic conditions (Hopkins and Pettingill 2017). That voters are casting ballots for mayors in a way that does not necessarily channel national political trends suggests that mayoral representation may be based on local, not national, politics. In a finding more relevant to this project, furthermore, they write that "in cities with their own TV stations and newspapers, there is a robust relationship between city-level unemployment and the [electoral] performance of the incumbent mayor" (Hopkins and Pettingill 2017, 710). That information about local economic conditions changes political behavior, particularly relative to national economic conditions, suggests that mayoral politics remains more focused on local issues.

GOVERNORS, MAYORS, AND NATIONALIZED POLITICAL SPEECH

The existing literature measures nationalization by examining the correlation between partisan vote shares at various levels of the federal electoral system; this is sensible, as elections allow citizens to hold elected officials accountable for their actions in office (Fearon 1999; Ferejohn 1986; Fiorina 1981), ensure proper representation (Verba, Schlozmann, and Brady 1995), and provide an avenue for expressive political participation (Hamlin and Jennings 2011). Yet while this approach has been useful in establishing the rise of nationalized political competition, it necessarily focuses on *voter* behavior just as much—if not more—than *elite* behavior. Because we are interested here in the nationalized behavior of political elites, we need a measure that is less dependent upon the actions of the mass public. For this reason, we measure nationalization by examining the similarity in mayoral and gubernatorial speech with that of members of Congress, captured via Twitter. The more a mayor or governor's rhetoric resembles that of members of Congress, the more nationalized we deem them to be.

Why might politicians use social media? A politician's social media content is a component of their strategic communication plan to engage with constituents (as well as a broader audience) directly, as well as indirectly through mediated messages. In this sense, Twitter is similar to press releases, news stories, or interviews. Twitter is a particularly important medium for politicians, as Twitter followers are likely to be opinion leaders within their districts—as well as journalists, activists, and local politicians—who convey what they read to a broader audience of constituents.[2] In the same way that we understand press releases and campaign speeches as providing useful data on elite communication to constituents, we expect politicians to use Twitter to cultivate constituent support and signal their priorities as lawmakers (Grimmer, Messing, and Westwood, 2012). Importantly for our purposes, social media provides us with an opportunity to capture and record a component of this strategic communication plan.

That communication is a key component of strategic elite-level behavior in modern politics has been shown by numerous scholars. One study, for instance, found that Republican officials tend to use veiled religious language that appeals to White evangelical Protestants. This "GOP code" is designed to both signify a politician's in-group partisan membership and solidify the support of sympathetic voters (Calfano and Djupe 2009). In addition, research has shown that politicians are increasingly and deliberately engaging in speech that seeks to outrage and belittle the opposing political party and its supporters. This verbal strategy, which Grimmer (2013) refers to as "partisan taunting," is yet another way that political elites seek to signal their in-group membership to their supporters in the electorate.

In addition to being used to signal in-group membership and taunt political opponents, politicians strategically speak about those issues and policies where their party is traditionally seen by the electorate as being more competent than the opposing party. Indeed, voters stereotypically assign issue positions to Democratic and Republican elites (Goggin and Theodoridis 2017). The assignment of these stereotypes largely occurs as result of the notion of "issue ownership," where one party is so deeply—and durably—focused on a particular issue that they are viewed by the electorate as "owning" that issue

2. Approximately 22% of US adults use Twitter, but Twitter users are younger, more educated and more likely to be Democrats than the general public, according to the Pew Internet and American Life Project (Wojcik and Hughes 2019).

(see, e.g., Petrocik 1996). By "repeatedly and consistently addressing problems stemming from [the owned] issues over time," candidates (and their associated party) can gain an electoral advantage (Banda 2016, 653).

Speech, then, has been shown to be an important aspect of politicians' presentational style. We extend this logic by arguing that, if they are experiencing the forces of nationalization, mayors and governors should be using rhetoric and patterns of speech that are similar to those of their congressional counterparts. Thus, if these more locally focused offices are nationalized, mayors and governors will be discussing the same topics as members of Congress.

While analyzing nationalization via elite speech on Twitter seems appealing, one potential problem with this approach is that citizens are not uniformly on Twitter. In fact, most citizens are not active on the social networking platform. Accordingly, it is possible that mayors and governors are engaging in nationalized rhetoric, but no one observes this. However, such a concern appears to be unfounded. Studies have shown that those individuals who are not on Twitter are nevertheless exposed to politicians' tweets via journalists' coverage of political affairs. Indeed, because journalists view tweets as newsworthy, they often quote from politicians' tweets—or link directly to the tweet itself—when writing their articles (Lawrence et al. 2014; McGregor and Molyneux 2018). Accordingly, it is possible for citizens who are not themselves users of Twitter to still come across tweets from their political representatives.

Another potential concern pertains to the trade-off between rhetoric and policy focus. While using nationalized rhetoric does not necessarily inhibit governors and mayors from engaging in locally focused policy, there are still costs associated with a politician's decision to use rhetoric that mirrors that of congressional officials. Indeed, one of the most important aspects of politicians' careers pertains to the ways in which they present themselves to the electorate (Fenno 1978). Adopting a presentational style that focuses largely on national politics will lead voters to assume that the politician cares largely about national issues. Focusing on local issues, by contrast, will lead voters to assume that a politician is most concerned with issues of local importance. Regardless of actual policy output and legislative focus, how politicians choose to present themselves to the electorate is a decision of great consequence.

As a measure of nationalization, we expect governors to be engaging in rhetoric—and discussing topics—similarly to members of Congress. Such an expectation comports with the existing literature, which argues that governors have become nationalized along both of the dimensions described above. Indeed, governors are often evaluated by the electorate through the lens of national politics, and gubernatorial elections are increasingly fought-over national issues. In addition to changes in the media landscape, one potential reason to expect governors to engage in nationalized rhetoric is that they often have ambitions for higher office. Governors are frequently candidates for president of the United States, and 17 presidents have previously held a state's governorship (Shapiro and Lawrence 2015). Engaging in nationalized rhetoric, then, is a potential way for governors to engage with issues of broad importance and build their national brand in preparation for a presidential bid.

In addition to their tendency to harbor ambitions for higher office, we expect governors to use nationalized rhetoric as a result of the nature of intraparty federalism. As Hopkins (2018) notes, the demise of the patronage era made state political parties increasingly dependent upon national party organizations for money and support. Such a top-down organizational structure changes the incentive structure for both state parties and state-level politicians, forcing a more national focus over more a regional approach. Such a shift has been exacerbated by the end of the solidly Democratic South and the polarization of state legislatures, two trends which have made state politics more similar to national political competition (Hopkins 2018; Shor and McCarty 2011).

There is less of an ex ante reason to assume that mayors engage in nationalized rhetoric. In fact, mayors may be qualitatively different than statewide and, in particular, national politicians. For one, mayors tend to focus on evaluating what can be built (and where), on public goods such as streets and sanitation, and on public safety (Oliver 2012). Faced with these daily—and immediate—tasks, mayors may simply have little time or energy to engage in rhetoric that mirrors that of members of Congress. Second, mayoral electoral campaigns often lack the intrigue and interest of those for higher levels of elected office. As Hopkins (2018, 61) notes, American politics "has become a contest of evocative symbols, and state and *local politics* are largely devoid of such symbols" (emphasis added). Absent the intrigue and "evocative symbols" that characterize races for federal or statewide offices, mayoral rhetoric should be expected to focus on issues of local importance. Similarly, mayoral elections do not necessarily occur at the same time as national campaigns and thus may not benefit from the media-market trends that align gubernatorial campaigns with national issues. Lacking these electoral incentives, mayors may be more likely to abjure nationalized rhetoric and focus their speech on topics of local importance.

Nevertheless, there are compelling pieces of evidence to indicate that mayoral representation might exhibit a degree of nationalization. First, recent presidential runs by mayors—including Pete Buttigieg of South Bend, IN, and Rudy Giuliani of New York City—suggest that some mayors, like governors, increasingly have their sights set on higher office. To the extent mayors harbor ambitions of obtaining higher office,

developing a national platform is essential. Engaging in nationalized rhetoric is one fruitful way to do this. Second, recent work suggests that mayoral behavior can and does align with national political agendas. Indeed, mayoral fiscal preferences are known to align with their partisan labels (Einstein and Glick 2018), and partisan control of local government is predictive of local government spending at both the county level (Choi et al. 2010; de Benedictis-Kessner and Warshaw 2020; Percival, Johnson, and Neiman 2009; Ybarra and Krebs 2010) and the mayoral level (de Benedictis-Kessner and Warshaw 2016, 2020; Einstein and Kogan 2015; Tausanovitch and Warshaw 2014).[3] To the extent partisanship is predictive of these local outcomes, and to the degree nationalization correlates with partisanship, we might expect mayors to engage in rhetoric—and discuss topics—similar to those of members of Congress.

Accordingly, understanding whether mayors engage in nationalized rhetoric and the situations under which they do so are important—and unanswered—questions. We theorize that mayors' representational style is distinct from congressional representation and, moreover, that we will be able to evaluate these differences in representational style via elite speech on Twitter. However, we also expect that a mayor's likelihood of engaging in nationalized rhetoric will vary based on certain mayoral and city-level covariates.

It is possible that mayors who represent cities with a strong partisan tilt will be more likely to use nationalized political rhetoric, which is in accordance with Grimmer's (2013) theory of polarized elite rhetoric. Drawing on a similar logic, we expect that mayors who govern over a city that leans heavily toward one of the two major parties should feel less constrained in their ability to speak out on national topics. On the other hand, those mayors who govern over cities with a more heterogeneous partisan makeup should focus their rhetoric on more local topics. In addition, we should expect that a mayor who is "misaligned" with the partisan makeup of her city—that is, a Democrat (Republican) governing a city comprised largely of Republicans (Democrats)—should use less nationalized rhetoric than one whose partisan affiliation matches the partisan leanings of their city. Such an expectation is rooted in a mayor's desire for reelection. When governing over a city that leans toward the opposing political party, a "misaligned" mayor will avoid speaking on nationalized partisan issues for fear of antagonizing her local constituents. Broadly speaking, then, we expect that mayors will use nationalized rhetoric when the partisan makeup of their city allows them to do so.

Moreover, in line with the approach in de Benedictis-Kessner and Warshaw (2020), we also expect that mayors' use of nationalized rhetoric will depend upon the institutional and population features of their city. We expect mayors to be more nationalized in their rhetoric when their city employs a city manager, as city managers engage in more of the apolitical day-to-day activities of running a city. Freed from the requirements of such activities, mayors whose city employs a city manager should have more time to engage in nationalized political rhetoric while still ensuring that the day-to-day business of his city is addressed. Finally, we expect the degree to which mayors engage in nationalized rhetoric to be increasing with the size of their city's population. As a city's population increases, each individual becomes less likely to have any form of personal engagement with the mayor's office. As a result, mayors will have less of an incentive to discuss local and personal topics and will have a greater opportunity to focus on topics that mirror those being discussed by members of Congress.

DATA

In order to examine the extent to which the nationalization of American politics has affected gubernatorial and mayoral behavior, we collected Twitter data for mayors, governors, and members of the US House of Representatives. These data allow us to construct our main dependent variable: the extent to which a mayor or governor employs language that is similar to members of Congress. The more similar mayoral or gubernatorial rhetoric is to that of members of Congress, the more "nationalized" we consider them to be.

Note that this measure of nationalization is distinct from partisanship. Partisanship can—and likely does—operate in different manners at various levels of government. For instance, a mayor's position on safety in public parks may fall along partisan lines at the local level. However, if this mayor is discussing the issue of park safety and members of Congress are not, then the mayor will have a low score on our measure of nationalization. At the same time, a mayor may be discussing things that are *not* guided by partisanship at the local level. However, if these topics are similar to those discussed by members of Congress, this mayor will have a high score on our measure of nationalization.

To begin, we collected the official Twitter handle for every mayor who is a member in the United States Conference of Mayors as of May 2018.[4] Though mayors are not uniformly on Twitter, we obtained Twitter handles for 587 (this amounts to

3. Earlier work did not necessarily support this finding (Ferreira and Gyourko 2009; Gerber and Hopkins 2011).

4. The United States Conference of Mayors is the official nonpartisan organization of cities with populations of at least 30,000. We chose this sample because we anticipated that mayors of medium- to large-sized

42% of the mayors eligible for inclusion in our study).[5] To avoid any biases that may be introduced by infrequent Twitter users, we excluded from our data all mayors who posted fewer than 100 tweets. This left us with a sample of 370 mayors, amounting to 27% of all mayors in the United States Conference of Mayors. In late December 2018, we scraped the previous 3,200 tweets for each of these Twitter handles.[6] Our focus is on tweets that were posted by the user, so we excluded all retweets from our analyses. This process generated a total of 404,049 tweets from our sample of mayors. Because our aim is to compare mayoral rhetoric with more nationalized speech, we collected 102,670 tweets from 49 governors and 964,810 tweets from 475 members of Congress using an identical collection protocol.[7] In all cases, our tweets were collected while the mayor, governor, or member of Congress was in office.

For our sample of mayors, we also collected individual-level attributes. This includes mayors' gender, racial identification, and length of term in office; we add to this the percentage of the vote share they received in their most recent election, as well as each mayor's partisan affiliation. These data were obtained from a combination of mayors' campaign websites and newspaper biographies of the mayors or of the mayoral election, or inferred from the mayor's collection of endorsements or source of campaign funds. For example, a mayor who received endorsements from prominent national Democrats was classified as a Democrat. On the other hand, a mayor who received funding from conservative political action groups would be classified as a Republican. Though many mayoral offices in the United States are nonpartisan, the approach we use here allows us to obtain a reasonable estimate as to each mayor's partisan affiliation. Accordingly, our data—by construction—contain no nonpartisan mayors. Information on our coding of mayoral partisanship can be found in the appendix.

We also collected information on the institutional features of each city in our data set. This includes indicators for whether the city government operates with a mayor-council system or a council-manager system, as well as whether the city employs a city manager. In mayor-council systems, a city's mayor is elected separately from the city council and often acts as the city's chief executive officer. Council-manager systems, by contrast, select the city's mayor from the city council. Under the council-manager system of local government, the mayor is comparatively weak, and the day-to-day operations of the city are overseen by an appointed or elected city manager. Though all city manager forms of government contain a city manager, city managers are also occasionally used in mayor-council systems.

To these individual- and institutional-level variables we added information on the demographic composition of each city. Using data from the most recent American Community Survey estimates, we collected the percentage of male and female residents in each city. We also collected information on the percentage of Black and Hispanic residents, as well as the percentage of individuals who own or rent their homes. We added to this data proxy measures of each city's election results for the 2016 presidential election. These estimates are constructed by using county-level presidential election returns that are weighted by the percentage of the city that is in each county. A summary of the mayoral data can be seen in appendix table 1.

QUANTIFYING TEXT DATA

In order to compare political rhetoric across mayoral, gubernatorial, and congressional offices, we consider two distinct strategies for text representation—first, a representation based on the presence or absence of specific words/phrases (the vocabulary); and second, a topic-based representation. We describe each of these approaches below.

Word/phrase vector representation

Research over the last two decades in machine learning has shown that regularized linear models can be combined with simple representations of text documents (such as the presence or absence of words or the adjusted frequency with which words appear) to produce accurate out-of-sample predictions in both classification and regression settings. For our purposes, we first establish the set of all phrases that will be considered in our analyses. We begin by tokenizing our collection of tweets (this breaks a sentence or stream of text into tokens, usually words), removing stopwords (such as *the*, *and*, *or*), and lemmatizing the words (this combines inflected forms of the same word, such that *going*, *go*, and *went* all become *go*). After these preprocessing steps, we combine our collection of words into bigrams, which are sequences of two words that appear next to each other.[8] We only consider

cities have access to Facebook and Twitter and possibly even staff to help monitor these accounts.

5. We first collected official accounts, and if an official account was not available, we then used a campaign account.

6. This is the maximum number of tweets we were allowed to get because of Twitter's API limit. We were able to get 200 tweets in each request. For each of our targeted politicians, we continued sending requests until we reached the 3,200 limit or their total tweet limit.

7. The governor of Alaska, Mike Dunleavy, has a private Twitter account that we were unable to scrape. We are otherwise able to collect all gubernatorial Twitter data.

8. This preserves some information about the relative positions between adjacent words and has been shown to improve the performance of

bigrams that appear in at least three different documents—that is, an individual's entire set of tweets—in the data. Finally, we reweight each document according to the term frequency inverse document frequency, a process that is analogous to weighting by the number of times that a bigram appears in the document divided by the number of documents in which that bigram appears at least once.

Topic representation

In addition to our word and phrase vector representation, we also consider the specific topics that mayors, governors, and members of Congress discuss. Topic models are typically generative models, where the text generating process is assumed to be that each document has a particular distribution over topics, and each topic has a distribution over words. In our case, "documents" refer to aggregated Twitter feeds and "words" refer to single words in a tweet. A document is generated by repeatedly first sampling a topic from its topic distribution, then sampling a word from that topic's word distribution (note that this loses sequence information, but is consistent with the "bag of words" style of document modeling). We use latent Dirichlet allocation (LDA) to learn topic models.[9]

We use regularized logistic regression for both our word/phrase vector representation analysis and our topic representation analysis. While simple, logistic regression has had tremendous success in text analytics. Here, we use logistic regression to predict whether a Twitter user is a mayor or member of Congress based on the user's tweets. As logistic regression models dichotomous outcomes, the result of our models can be interpreted as the probability that user i is a member of Congress or not (1 = yes, 0 = no). We apply a L2 penalty on our analysis to guard against overfitting our models. A more detailed technical description of our approach can be found in the appendix.

EMPIRICAL ANALYSES

Our main empirical strategy is to use quantifications of the text produced by different types of political actors to analyze how similar they are to each other in terms of their Twitter rhetoric. Recall that we define a mayor or a governor as being more "national" the more closely their rhetoric resembles that of members of Congress. Our first set of analyses focuses on measuring differences between these actors in topic space, while the second set leverages the ease or difficulty of predicting individual membership in a particular group as a measure of how similar two groups are.

Topic differences: Mayors, governors, and members of Congress

We start our analysis with a simple validity check on the plausibility of our hypotheses. We separately infer topic models using LDA (Blei, Ng, and Jordan 2003) on members of Congress' tweets, governors' tweets, and mayors' tweets and visually inspect the words associated with all topics. We list some of these for illustration in table 1.[10] The columns in table 1 denote various offices, while the rows indicate unique topics generated by our LDA model. The generation of these topics was calculated without any partisan consideration—that is, we did not separately infer LDA topic models for Democratic and Republican officials. We can see that, at least to a quick human inspection, the congressional twitter topics are more "national" than those of mayors. Mayoral topics include terms such as "park," "library," "police," and "street"; Congressional topics include terms such as "American," "tax," "national," and "congress." Gubernatorial topics more closely resemble those of members of Congress.[11]

Differences in topic distributions: Direct measurement. We now turn to a more systematic investigation of topic differences. Our first strategy is to calculate average distances between all the individuals in our data set in topic space and then compare the distributions of differences between different types of individuals (where the types are mayors, governors, and members of Congress).

Because the LDA training process is known to be unstable (Roberts, Stewart, and Tingley 2016), we first train 100 different topic models (based on different random initializations) and then represent each individual according to each of these 100 topic models. Thus, our use of the term "distance" refers to the average of the 100 different distances computed between topics for two individuals in this manner. The distance itself is the Jensen-Shannon divergence, which simply indicates how distinguishable one distribution is from another. This measure has an upper limit of 1, which would indicate a complete lack of overlap between two distributions. For our purposes, we are attempting to determine whether the distribution of mayoral, gubernatorial, and congressional twitter topics can be distinguished from each other. Greater distances—that is, higher

classification models over models that use single words only in multiple sentiment analysis tasks (Wang and Manning 2012).

9. Structural topic models (Roberts et al. 2013) incorporate covariates (e.g., time, author) of the source text data in topic models. However, we are using the representation in the topic space to create a useful measure of distance. In this case, we do not want to allow the model to condition on these covariates, since they may contribute to the distances. Therefore we use a standard LDA model, which only uses text data for training.

10. Using 26 topics.

11. Full word lists by topic are available in app. tables 4–6.

Table 1. Congress versus Governor versus Mayor Tweet Topics

Congress	Governor	Mayor
American, people, must,	state, emergency, stay,	center, opening, new,
Congress, Republican, keep,	local, road, storm,	grand, ribbon, summer,
need, want, fight, . . .	county, update, weather, . . .	bike, north, green, . . .
president, Trump, administration,	school, teacher, power,	police, fire, officer,
statement, Obama, decision,	early, vote, election,	department, chief, step,
court, policy, judge	voice, student, help, . . .	year, law, crime, . . .
tax, job, business,	job, new, economic,	street, tax, water,
small, cut, worker,	company, create, development,	road, power, south,
economy, new, energy, . . .	economy, business, investment, . . .	crew, main, traffic, . . .
health, care, access,	justice, reform, need,	new, park, check,
plan, million, insurance,	Republican, president, house,	coming, downtown, art,
affordable, cost, coverage, . . .	Trump, federal, criminal, . . .	building, exciting, dog
national, answer, threat,	thank, everyone, hope,	free, office, public,
force, asked, concern,	helping, great, volunteer,	city, library, park,
new, release, official, . . .	food, day, hosting, . . .	information, info, monday

numerical values—indicate more easily distinguishable topic distributions.[12]

Another point worth noting is that we must adjust for frequency of tweeting. Training LDA models directly on all data from all individuals would bias the topics learned toward active users. There is no standard methodology yet for directly comparing documents of different lengths (Gong et al. 2018), so we employ the simple approach of randomly subsampling 100 tweets from each of the individuals and concatenating these 100 tweets into a single document representing each individual.[13]

The results of this analysis are presented in figure 1. Some insights stand out immediately simply from visual inspection of the figure. First, the distributions of differences within each type of individual are much tighter, with smaller means and medians, than the distribution of differences between members of Congress and mayors, which exhibits a much heavier tail of high distances. Second, there is a similar phenomenon even when comparing the distribution of differences between (1) members of Congress and governors and (2) members of Congress and mayors. The former distribution is tighter, with a lower mean and median and much thinner tail.[14]

These results suggest that members of Congress and mayors systematically focus their Twitter rhetoric on different topics. Further, the finding that the distance between governors and members of Congress is smaller than the distance between mayors and members of Congress indicates that governors are likely using more nationalized rhetoric than mayors. Of course, this does not mean that mayoral use of Twitter is devoid of nationalized rhetoric. Indeed, a comparison with a truly apolitical group (Premier League soccer players) shows that, as one would expect, the distance between soccer players and members of Congress is greater than that between mayors and members of Congress (see the appendix for details). This suggests that mayoral rhetoric does, in fact, exhibit a degree (albeit a moderate one) of nationalization—this is to be expected, as the mayoral group and the congressional group are both producing political speech.

Differences in topic distributions: Predictability of type. The analysis above examines differences in topic distributions by directly measuring these differences. We can approach the problem in another way as well, by asking whether choice of topics to tweet about allows us to reliably distinguish mayors and members of Congress. The simplest way to understand our test is as follows: given (anonymous) information on only the distribution of topics a Twitter user tweets about, can we distinguish whether that Twitter user is a member of

12. The Jensen-Shannon divergence is a symmetrized and smoothed version of the Kullback-Leibler divergence. The Kullback-Leibler divergence is a frequently used measure to determine the distance between probability distributions.

13. An implementation note: we use the Gensim package for LDA (Řehůřek and Sojka 2010) with 26 topics (based on optimizing the cross-validated coherence score—see the appendix for more details). Each model is trained for 45 epochs. All other settings are the package defaults.

14. Paired *t*-tests show that the mean of the difference in distances is significantly higher for the comparison between members of Congress and mayors than for any of the other four distributions, and two-sample Kolmogorov-Smirnov test results also indicate that the distribution of differences between members of Congress and mayors is statistically distinguishable from each of the others using 95% confidence intervals.

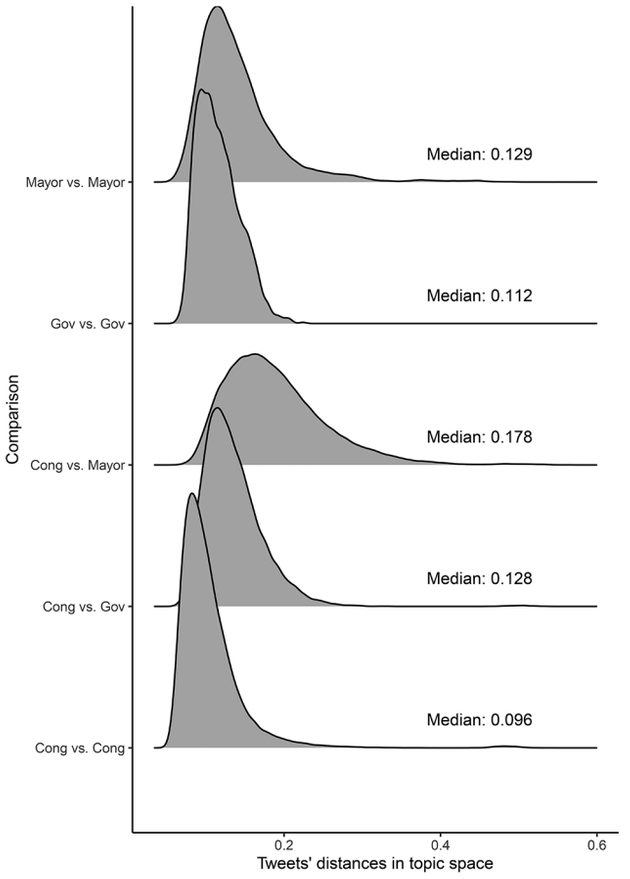


Figure 1. Distribution of topic distances

Congress or a mayor? If we can, this strengthens our case that mayors and members of Congress use different rhetorical strategies in communicating on Twitter.

We adopt a standard machine learning approach for this test. We use the 100 different topic models described previously to first represent each member of Congress and each mayor in the same way as above. Thus, in each model, each individual is represented as a 26-dimensional feature vector $\mathbf{x}_i$ paired with a label $y_i \in \{0, 1\}$. Each element of $\mathbf{x}_i$ reflects the proportion of user i's Twitter language assigned by the model to that specific topic. The variable y_i is 0 if individual i is a mayor and 1 if a member of Congress. For each of the 100 topic models, we run fivefold cross-validation to perform out-of-sample classification for each individual. Cross-validation operates by randomly permuting the data, separating them into k (in our case $k = 5$) "folds" and then repeatedly training a classification model on $k - 1$ of the folds and testing on the remaining one. In our case, we simply use logistic regression to learn from the four folds each time.

We measure the performance of the classifier using the area under the ROC curve (AUC). The ROC curve, or receiver operating characteristics curve, plots the fraction of correctly specified observations against incorrectly specified observations (e.g., Twitter users correctly specified as mayors or incorrectly specified as mayors).[15] The AUC, then, is a measure of how accurate our classification scheme is; this measure ranges from 0 to 1, with higher values indicating a greater predictive accuracy. The advantage of the AUC over directly measuring accuracy is that it is agnostic to the proportions of the two groups in the data and comparable across settings where the data set is differently balanced. The higher the AUC score, the better we are able to correct classify Twitter users as a mayors (as opposed to members of Congress) based on the topics about which they choose to tweet.

We find that the average AUC over the 100 different topic models is .9849 $\pm$ .0004. This is stunningly high. It shows that mayors can be very easily distinguished (out of sample) from members of Congress simply by comparing topic distributions. It further demonstrates the gap between mayoral and congressional rhetoric on Twitter.[16]

Variation in nationalized mayoral rhetoric

Measuring similarity to congressional speech: The congressional similarity score. While the preceding analyses suggest that gubernatorial rhetoric closely matches that of members of Congress, mayors appear to be using a different rhetorical style. However, these results do not indicate that *all* mayors are similarly focused on local issues. In fact, it is possible that some mayors use language more similar to congressional and gubernatorial rhetoric than others. Examining the sources of variation in nationalized mayoral rhetoric is the task to which we now turn.

We start by using the insights from the predictability of type (mayor or member of Congress) to define a precise measure of nationalized rhetoric as expressed by mayors. We call this measure the congressional similarity (CS) score. We again use the machine learning task of learning a classifier to distinguish between members of Congress and mayors. Our intuition behind this process is that mayors who score closer to members of Congress are more similar to them and therefore are more likely to be nationalized in their use of rhetoric.

After training an LDA model and representing each mayor and member of Congress in the topic space, for each mayor i, we train a logistic regression classifier based on the documents corresponding to all other mayors (labeled as 0) and members of Congress (labeled as 1). We then predict the probability that

15. That is, we are plotting the true positive rate on the Y-axis and the false positive rate on the X-axis.

16. We apply DeLong's method to compute the standard deviation of the AUC (DeLong, DeLong, and Clarke-Pearson 1988).

mayor *i* is a member of Congress using this leave-one-out model (importantly, the mayor is not in the training set used to build the logistic scoring function) and consider that prediction the CS score.[17]

Association between congressional similarity and covariates. To begin to understand the sources of variation in nationalized mayoral rhetoric, we looked for associations between the degree of nationalized mayoral rhetoric and covariates collected about mayors, their city's institutions, and the constituents they represent. As our metric of mayoral nationalization, we use our measure of similarity to congressional rhetoric—the CS score—defined above. The greater the CS score, the more similar the mayor's rhetoric is to the more nationalized political speech of members of Congress.

To examine the sources of heterogeneity in mayoral rhetoric, we regress the CS score on indicator variables for institutions (e.g., city manager, council-manager system), the total number of years the mayor has served in office, the city population (rescaled in units of 100,000), and the vote share won by the mayor in the previous election.[18] We present models with and without additional covariates—we can also control (among other things) for mayoral race, gender, and party—in table 2. We also control for city-level partisanship in two different ways. In one operationalization, city partisanship is captured by the city-level percentage of the Democratic presidential vote in 2016. In the second measure, we calculate the absolute value of the difference of this measure from 50. Higher values on this measure indicate a city that is highly Democratic or Republican. Thus, while the former measure captures the raw city-level partisanship, the second measure captures city-level partisan extremity. We also present results from models that contain additional city-level covariates. In one specification, we include the percentage of the city that is male, the percentage of the city that is White, and the percentage of the city that is Black. In a final specification, we add the percentage of residents who own their home. Though the empirical strategy we use here does not allow us to make any causal claims, the relationships we show nevertheless help us better understand the sources of variation in nationalized mayoral rhetoric.

As shown in table 2, we find that the number of years a mayor has served in office is negatively associated with nationalized rhetoric. This suggests that longer-serving mayors are more focused on their citizens' local needs than their counterparts who have less experience in office. In addition, we find a positive relationship between our city population variable and nationalized rhetoric. This indicates that, as a city's population increases, mayoral speech more closely mirrors that of a congressional representative. This lends support to our theoretical expectation that, because citizens of larger cities are less likely to experience any form of personal contact with their mayor, mayors can adopt a more nationalized representational style. Because this relationship persists even when we control for city-level sociodemographic factors, this finding is likely driven by something that is inherently different about larger cities and *not* by some factor that happens to correlate with large cities. Finally, we find no statistically significant relationship between operating under a council-manager system and nationalized rhetoric, nor do we find evidence of a relationship between employing a city manager and nationalized rhetoric.

Similarly, we find no statistically significant relationship between city-level partisanship and nationalized mayoral rhetoric. Neither the city's Democratic presidential vote share in 2016 nor its partisan extremity are associated with a greater amount of nationalized mayoral speech. However, recall from above that part of our theoretical expectation is that mayors who govern over a city that is not in line with their own partisan leaning will use less nationalized rhetoric than those whose partisan affiliation matches the partisan leaning of their city. To examine whether this is the case, we reestimated the model shown in table 2, column 7, but included a dummy variable indicating whether or not a mayor is "misaligned." This variable takes on a value of 1 if the mayor is a Republican (Democrat) governing a city that gave more (less) than 50% of its vote share to Hillary Clinton in 2016 and 0 otherwise. In no case does this variable reach any conventional level of statistical significance. Moreover, we find no partisan differences in terms of mayoral misalignment and the use of nationalized rhetoric. These abridged results are shown in table 3; the full table can be found in appendix table 2.[19]

While the results presented above should be interpreted with caution, as we are limited to a small set of mayors, it is encouraging that we observe similar results regardless of the

17. The following mayoral Twitter IDs have topic distributions most similar to members of Congress: mayorcorymason, patownhall, joshfryday, mayorcooper, and edsachs. Those who are most different include: mayorlozanobp, atchison4mayor, mayormcleod, bethlehemmayor, ananforoakpark.

18. In cases where a city's mayor is appointed to the position by the city council, we treat the mayor's vote share in the previous election as the percentage of the vote they received in their most recent election to the city council.

19. Because our measure of misalignment is constructed using a city's Democratic vote share, we do not condition on city partisanship in these models. However, the results do not change upon the inclusion of either of our two measures of city partisanship.

Table 2. Coefficients for Congressional Similarity

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Years in office	−.003**	−.003***	−.003***	−.003***	−.003***	−.003***	−.003***
	(.001)	(.001)	(.001)	(.001)	(.001)	(.001)	(.001)
City population	.008***	.007***	.007***	.006***	.006***	.006***	.006***
	(.002)	(.002)	(.002)	(.002)	(.002)	(.002)	(.002)
Previous vote share	−.0001	−.0001	−.0001	−.0001	−.0001	−.0001	−.0001
	(.0003)	(.0003)	(.0003)	(.0003)	(.0003)	(.0003)	(.0003)
Council-manager system	−.031	−.024	−.025	−.024	−.025	−.025	−.025
	(.020)	(.019)	(.019)	(.020)	(.020)	(.020)	(.020)
City manager	.024	.025	.026	.027	.027	.029	.029
	(.020)	(.020)	(.019)	(.020)	(.020)	(.020)	(.020)
White		.007	.006	.014	.014	.014	.014
		(.013)	(.013)	(.014)	(.014)	(.014)	(.014)
Male		.033***	.034***	.033***	.033***	.031**	.032***
		(.012)	(.012)	(.012)	(.012)	(.012)	(.012)
Republican		−.029**	−.034***	−.026**	−.032***	−.020	−.026**
		(.012)	(.011)	(.012)	(.011)	(.013)	(.012)
City partisanship:							
Democratic presidential share, 2016		.001		.001		.001	
		(.0004)		(.0004)		(.0004)	
Presidential share extremity			.001		.001		.001
			(.001)		(.001)		(.001)
% male (city)				.001	.001	−.0001	−.0003
				(.004)	(.004)	(.004)	(.004)
% White (city)				−.00001	−.0002	.0001	−.0001
				(.0005)	(.0004)	(.0005)	(.0004)
% Black (city)				.0005	.0003	.0004	.0003
				(.0005)	(.0005)	(.0005)	(.0005)
% own home (city)						−.001	−.001
						(.0004)	(.0004)
Constant	.264***	.217***	.239***	.157	.205	.238	.281
	(.019)	(.031)	(.023)	(.190)	(.186)	(.200)	(.196)
R^2	.079	.124	.124	.128	.129	.133	.132

Note. $N = 370$. Standard errors are in parentheses.
* $p < .1$.
** $p < .05$.
*** $p < .01$.

model specification. Among the set of variables measuring mayoral demographics, only gender and partisanship are meaningful predictors of using rhetoric that is similar to members of Congress. Though there are many potential reasons why these variables are associated with nationalized mayoral rhetoric, two possibilities seem likely. First, there are noted gender differences when it comes to the decision to run for political office. While women tend to run for office out of a desire to work on fixing problems, men appear to be more motivated by the pursuit of power (Schneider et al. 2015). This pursuit of power and of higher office could be driving men's use of great nationalized rhetoric relative to women. That Republican mayors tend to use less nationalized rhetoric than others likely results from the nature of Republican political thought. Indeed, because Republicanism tends to prioritize state and local control over a strong national government, Republican mayors are more likely to focus on the needs and issues facing their local communities. However, we stress that these are merely potential reasons for the relationship between gender, partisanship, and nationalized mayoral rhetoric. Moreover, we are cautious about overinterpreting the meaning behind these associations, as we did not have an ex ante theoretical reason to expect that a mayor's gender or partisan affiliation would alter their level of nationalized rhetoric. More precisely determining

Table 3. Misaligned Mayors and Nationalized Rhetoric: Congressional Similarity

	(1)	(2)	(3)
Misaligned mayor	−.001	.002	.003
	(.011)	(.011)	(.015)
Republican		−.027**	−.026*
		(.012)	(.015)
Misaligned mayor × Republican			−.003
			(.023)
Constant	.322	.291	.293
Controls?	(.197)	(.197)	(.198)
	✓	✓	✓
R^2	.116	.128	.128

Note. $N = 370$. Standard errors are in parentheses.
* $p < .1$.
** $p < .05$.
*** $p < .01$.

the theoretical linkages between these variables is something that we leave for future research.

CONCLUSION AND DISCUSSION

Among the deleterious consequences of the nationalization of American politics is the concern that elites' focus on national issues comes at the expense of attending to more pressing, locally based problems. The results we have presented suggest that, while there is variation, mayors have largely remained focused on local issues and use less nationalized speech than governors and members of Congress. The degree to which citizens' preferences and needs are represented by their elected officials is an important barometer of the health of a democratic society. In the case of the United States, which has experienced a considerable amount of political nationalization over the past few decades (Hopkins 2018), this metric has trended in a negative direction at the federal and gubernatorial levels. However, our analyses suggest that there is room for optimism about the vibrancy of representation provided by mayors. Unlike elected officials at other levels of government, the average American mayor currently appears to be unaffected by the nationalization of American politics. On the contrary, mayoral rhetoric tends to remain focused on the needs and concerns of the local citizenry. This finding persists even when controlling for a multitude of mayoral- and city-level covariates.

However, our results also suggest that mayors do not uniformly focus on local issues. Nationalized mayoral rhetoric is more likely to occur among mayors who govern cities with a large population. Such a finding likely arises because a large municipal population diminishes the likelihood that any citizen has a personal experience or point of contact with her mayor. As a result, mayors are freer to engage in national rhetoric and abjure local issues. Though plausible, such an explanation is only one potential reason that we observe a relationship between city size and nationalized mayoral rhetoric. Future work should more thoroughly examine the mechanism linking a city's population to mayoral representational style.

One final concern pertains to the durability of these results. The nationalization of American elections did not suddenly develop and alter congressional and gubernatorial behavior. In reality, the nationalization of American elections has been a secular process that has resulted from the complex interplay of ideological realignment, partisan sorting, and a changing media environment. Whether these trends in nationalization will eventually affect mayors, or whether Americans' most immediate chief executives will remain focused on their local mandate, is a question that can only be answered in time.

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